DML Commands in MySQL – Exercise 3

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1. USE A DATABASE

   USE CollegeDB;

   Output:

```
Database changed
```

2. CREATE THE TABLES (structure from Exercise 1)

```
CREATE TABLE Student (RegNo INT PRIMARY KEY, Name VARCHAR(15), Gender CHAR(1), DOB
DATE, MobileNo BIGINT, City VARCHAR(15));

CREATE TABLE Faculty (FacNo VARCHAR(4) PRIMARY KEY, FacName VARCHAR(15), Gender
CHAR(1), DOB DATE, DOJ DATE, MobileNo BIGINT, Dept VARCHAR(10));

CREATE TABLE Department (DeptNo VARCHAR(4) PRIMARY KEY, DeptName VARCHAR(15),
DeptHead VARCHAR(4));

CREATE TABLE Course (CourseNo VARCHAR(3) PRIMARY KEY, CourseDesc VARCHAR(14),
CourseType CHAR(1), SemNo CHAR(1), HallNo VARCHAR(4), FacNo VARCHAR(4));

CREATE TABLE Score (RegNo INT, CourseNo VARCHAR(3), Score INT);
```

   Output:

```
Query OK, 0 rows affected (0.167 sec)

Query OK, 0 rows affected (0.085 sec)

Query OK, 0 rows affected (0.191 sec)

Query OK, 0 rows affected (0.155 sec)

Query OK, 0 rows affected (0.074 sec)
```

3. SHOW THE TABLES

   SHOW TABLES;

   Output:

```
+---------------------+
| Tables_in_collegedb |
+---------------------+
| course              |
| department          |
| faculty             |
| score               |
| student             |
+---------------------+
5 rows in set (0.054 sec)
```

Question 1 – Populate all the five tables with your own data

4. INSERT RECORDS INTO STUDENT (all columns)

```
INSERT INTO Student VALUES
(191711341, 'Arun', 'M', '2003-05-14', 9876501234, 'Chennai'),
(191711342, 'Ramesh', 'M', '2002-11-02', 9876505678, 'Salem'),
(191711343, 'Priya', 'F', '2003-02-20', 9876509012, 'Madurai'),
(191711344, 'Ashwin', 'M', '2003-08-09', 9876503456, 'Coimbatore'),
(191711345, 'Divya', 'F', '2002-12-25', 9876507890, 'Salem'),
(191711346, 'Kamali', 'F', '2003-07-17', 9876504321, 'Trichy');
```

Output:

```
Query OK, 6 rows affected (0.014 sec)
Records: 6  Duplicates: 0  Warnings: 0
```

5. INSERT RECORDS INTO FACULTY

```
INSERT INTO Faculty VALUES
('F001', 'Ratnam', 'M', '1975-04-12', '2010-06-15', 9840012345, 'CSE'),
('F002', 'Kamal', 'M', '1980-09-23', '2012-07-02', 9884023456, 'ECE'),
('F003', 'Meena', 'F', '1985-01-30', '2014-11-10', 9444034567, 'IT'),
('F004', 'Mohan', 'M', '1970-12-05', '2008-06-20', 9790045678, 'MECH'),
('F005', 'Lakshmi', 'F', '1982-03-18', '2016-01-04', 9962056789, 'CIVIL');
```

Output:

```
Query OK, 5 rows affected (0.008 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

6. INSERT RECORDS INTO DEPARTMENT

```
INSERT INTO Department VALUES
('D001', 'CSE', 'F001'),
('D002', 'ECE', 'F002'),
('D003', 'IT', 'F003'),
('D004', 'MECH', 'F004'),
('D005', 'CIVIL', 'F005');
```

Output:

```
Query OK, 5 rows affected (0.013 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

7. INSERT RECORDS INTO COURSE

```
INSERT INTO Course VALUES
('C001', 'Statistics', 'T', '1', 'H001', 'F001'),
('C002', 'Programming', 'P', '2', 'H002', 'F002'),
('C003', 'Networks', 'T', '3', 'H003', 'F003'),
('C004', 'Databases', 'P', '4', 'H002', 'F001'),
('C005', 'Thermodynamics', 'T', '5', 'H004', 'F004');
```

Output:

```
Query OK, 5 rows affected (0.010 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

8. INSERT RECORDS INTO SCORE

```
INSERT INTO Score VALUES
(191711341, 'C001', 78),
(191711341, 'C002', 85),
(191711342, 'C001', 66),
(191711342, 'C002', 72),
(191711343, 'C003', 90),
(191711344, 'C002', 55),
(191711345, 'C004', 81),
(191711346, 'C001', 74),
(191711346, 'C005', 62);
```

Output:

```
Query OK, 9 rows affected (0.049 sec)
Records: 9  Duplicates: 0  Warnings: 0
```

9. INSERT USING A COLUMN LIST (one more student)

```
INSERT INTO Student (RegNo, Name, Gender, DOB, MobileNo, City)
VALUES (191711347, 'Harini', 'F', '2003-03-03', 9876502222, 'Erode');
```

Output:

```
Query OK, 1 row affected (0.051 sec)
```

Question 2 – View all the records from the five tables

10. VIEW ALL RECORDS FROM STUDENT
    SELECT * FROM Student;

    Output:

```
+-----------+--------+--------+------------+------------+------------+
| RegNo     | Name   | Gender | DOB        | MobileNo   | City       |
+-----------+--------+--------+------------+------------+------------+
| 191711341 | Arun   | M      | 2003-05-14 | 9876501234 | Chennai    |
| 191711342 | Ramesh | M      | 2002-11-02 | 9876505678 | Salem      |
| 191711343 | Priya  | F      | 2003-02-20 | 9876509012 | Madurai    |
| 191711344 | Ashwin | M      | 2003-08-09 | 9876503456 | Coimbatore |
| 191711345 | Divya  | F      | 2002-12-25 | 9876507890 | Salem      |
| 191711346 | Kamali | F      | 2003-07-17 | 9876504321 | Trichy     |
| 191711347 | Harini | F      | 2003-03-03 | 9876502222 | Erode      |
+-----------+--------+--------+------------+------------+------------+
7 rows in set (0.019 sec)
```

11. VIEW ALL RECORDS FROM FACULTY
    SELECT * FROM Faculty;

    Output:

```
+-------+---------+--------+------------+------------+------------+-------+
| FacNo | FacName | Gender | DOB        | DOJ        | MobileNo   | Dept  |
+-------+---------+--------+------------+------------+------------+-------+
| F001  | Ratnam  | M      | 1975-04-12 | 2010-06-15 | 9840012345 | CSE   |
| F002  | Kamal   | M      | 1980-09-23 | 2012-07-02 | 9884023456 | ECE   |
| F003  | Meena   | F      | 1985-01-30 | 2014-11-10 | 9444034567 | IT    |
| F004  | Mohan   | M      | 1970-12-05 | 2008-06-20 | 9790045678 | MECH  |
| F005  | Lakshmi | F      | 1982-03-18 | 2016-01-04 | 9962056789 | CIVIL |
+-------+---------+--------+------------+------------+------------+-------+
5 rows in set (0.013 sec)
```

12. VIEW ALL RECORDS FROM DEPARTMENT
    SELECT * FROM Department;

    Output:

```
+--------+----------+----------+
| DeptNo | DeptName | DeptHead |
+--------+----------+----------+
| D001   | CSE      | F001     |
| D002   | ECE      | F002     |
| D003   | IT       | F003     |
| D004   | MECH     | F004     |
| D005   | CIVIL    | F005     |
+--------+----------+----------+
5 rows in set (0.004 sec)
```

13. VIEW ALL RECORDS FROM COURSE

SELECT * FROM Course;

Output:

```
+----------+----------------+------------+-------+--------+-------+
| CourseNo | CourseDesc     | CourseType | SemNo | HallNo | FacNo |
+----------+----------------+------------+-------+--------+-------+
| C001     | Statistics     | T          | 1     | H001   | F001  |
| C002     | Programming    | P          | 2     | H002   | F002  |
| C003     | Networks       | T          | 3     | H003   | F003  |
| C004     | Databases      | P          | 4     | H002   | F001  |
| C005     | Thermodynamics | T          | 5     | H004   | F004  |
+----------+----------------+------------+-------+--------+-------+
5 rows in set (0.017 sec)
```

14. VIEW ALL RECORDS FROM SCORE

SELECT * FROM Score;

Output:

```
+-----------+----------+-------+
| RegNo     | CourseNo | Score |
+-----------+----------+-------+
| 191711341 | C001     |    78 |
| 191711341 | C002     |    85 |
| 191711342 | C001     |    66 |
| 191711342 | C002     |    72 |
| 191711343 | C003     |    90 |
| 191711344 | C002     |    55 |
| 191711345 | C004     |    81 |
| 191711346 | C001     |    74 |
| 191711346 | C005     |    62 |
+-----------+----------+-------+
9 rows in set (0.051 sec)
```

15. SELECT SPECIFIC COLUMNS

SELECT RegNo, Name, City FROM Student;

Output:

```
+-----------+--------+------------+
| RegNo     | Name   | City       |
+-----------+--------+------------+
| 191711341 | Arun   | Chennai    |
| 191711342 | Ramesh | Salem      |
| 191711343 | Priya  | Madurai    |
| 191711344 | Ashwin | Coimbatore |
| 191711345 | Divya  | Salem      |
| 191711346 | Kamali | Trichy     |
| 191711347 | Harini | Erode      |
+-----------+--------+------------+
7 rows in set (0.013 sec)
```